KENYA CERTIFICATE OF SECONDARY EDUCATION

COMPUTER PROJECT 2026 (451/3)

AZANI

INTERNET SERVICES PROVIDER

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YEAR OF EXAMINATION: 2026

SCHOOL: ST COLUMBANS HIGH SCHOOL

DECLARATION

STUDENT'S DECLARATION

I, **Makokha Viona Echesa**, a student at **ST COLUMBANS**, hereby declare that this project titled **" AZANI INTERNET SERVICES PROVIDER INFORMATION SYSTEM "** is my original work and has not been presented for any academic award or qualification in any other institution.

All references and materials used in the development of this project have been duly acknowledged. Any contributions or assistance from other individuals or sources have been appropriately cited and credited.

I understand that plagiarism is a serious academic offense, and I affirm that this work is a true reflection of my understanding, effort, and creativity.

Name: _____

Signature: _____

Date: _____

SUPERVISOR DECLARATION

This is to certify that the project titled **"AZANI INTERNET SERVICES PROVIDER INFORMATION SYSTEM"** has been submitted by **Makokha Viona Echesa**, of ST COLUMBANS HIGH SCHOOL, under my supervision.

I confirm that this project is the student's original work and has been developed under my guidance. The student has adhered to the required academic standards, and all sources of information used in the project have been appropriately acknowledged.

To the best of my knowledge, this work has not been submitted for any other academic award or qualification in any other institution.

Supervisor's Name: _____

Signature: _____

Date: _____

DEDICATION

This project is dedicated to:

My parents, for their unwavering support, encouragement, and sacrifices throughout my education.

My teachers, for their guidance, patience, and invaluable knowledge that have shaped my academic journey.

My friends and classmates, for their motivation and camaraderie during the challenging moments of this project.

The Almighty God, for granting me the strength, wisdom, and perseverance to complete this work.

Your belief in me has been my driving force, and I am deeply grateful for your constant presence in my life.

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ACKNOWLEDGMENT

I would like to express my sincere gratitude to all those who contributed to the successful completion of this project. This work would not have been possible without their support, guidance, and encouragement.

First and foremost, I am deeply thankful to my **Computer Studies teacher**, **[Teacher's Name]**, for their invaluable guidance, patience, and expertise throughout the development of this project. Their constructive feedback and constant encouragement kept me motivated and focused.

I am also grateful to the **school administration** of **[Your School Name]** for providing the necessary resources and a conducive environment for learning and innovation. Their support has been instrumental in the completion of this project.

Special thanks go to my **parents and family** for their unwavering support, encouragement, and sacrifices. Their belief in me has been a constant source of inspiration.

I would also like to acknowledge my **friends and classmates** for their moral support, collaboration, and encouragement during the challenging moments of this project. Their input and feedback were invaluable.

Lastly, I thank the **Almighty God** for granting me the strength, wisdom, and perseverance to complete this work.

To all who contributed directly or indirectly to this project, I am truly grateful.

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CHAPTER ONE: INTRODUCTION

1.1 Background Information

Azani Internet Services Provider is a Kenyan-based company dedicated to delivering reliable internet connectivity and infrastructure to learning institutions. These include primary schools, junior secondary schools, senior secondary schools, and colleges. The company's mission is to bridge the digital divide in education by offering affordable and scalable internet solutions tailored to institutional needs.

To access Azani's services, institutions must first pay a registration fee of **KSh 8,500**. After registration, Azani conducts a site visit to assess the institution's readiness for internet installation. This includes evaluating the number of users, availability of computers, and the presence of a Local Area Network (LAN). Institutions that meet the requirements pay an installation fee of **KSh 10,000**. Those lacking infrastructure may purchase personal computers and LAN nodes directly from Azani.

The monthly cost of internet services depends on the selected bandwidth, as shown below:

Bandwidth	*Monthly Cost (KSh)*
4 MBPS	1,200
10 MBPS	2,000
20 MBPS	3,500
25 MBPS	4,000
50 MBPS	7,000

Institutions upgrading to higher bandwidth receive a **10% discount** on the new package. Monthly payments are due by the end of each month. Late payments attract a **15% fine**, and services are disconnected if dues remain unpaid by the **10th of the following month**. Reconnection requires payment of the outstanding amount plus a **KSh 1,000 reconnection fee**

CHAPTER TWO: SYSTEM ANALYSIS

2.1 Problem Definition

Azani Internet Services Provider delivers internet connectivity and infrastructure to learning institutions such as primary schools, junior secondary schools, senior secondary schools, and colleges. However, the current manual approach to managing registrations, payments, infrastructure assessments, and service monitoring presents several challenges:

Inefficient Institution Registration

- Details of institutions and their contact persons are recorded manually.
- This process is time-consuming, prone to errors, and makes retrieval of information difficult.

Difficulty in Tracking Service Changes

- Institutions may upgrade bandwidth or purchase additional infrastructure.
- Manual updates often lead to inconsistencies and delays in reflecting changes.

Financial Management Challenges

- Payments for registration, installation, monthly subscriptions, fines, and reconnection fees are tracked manually.
- This leads to errors in calculations, delays in reporting, and difficulties in monitoring financial performance.

Lack of Centralized Data Storage

- Institution details, payment records, and infrastructure requirements are stored in separate files.
- This fragmentation makes it difficult to access and analyze data efficiently.

Limited Reporting Capabilities

- **Generating reports on registered institutions, defaulters, disconnections, and infrastructure requirements is tedious.**
- **Management often lacks timely and accurate reports for decision-making.**

Dependency on Manual Processes

- The entire process relies heavily on manual work, which is inefficient and unsustainable as the number of institutions grows.

These challenges highlight the need for an automated system that can streamline operations, improve data management, and provide accurate and timely reports. The **Azani Internet Service Provider Information System** aims to address these problems by offering a centralized, efficient, and user-friendly solution.

2.2 Overview of the Existing System

Currently, Azani manages its operations manually using paper records, spreadsheets, and physical documentation.

- **Registration**: Institutions pay a registration fee of KSh 8,500, recorded manually.
- **Installation**: Readiness checks and installation fees (KSh 10,000) are tracked separately.
- **Infrastructure Sales**: Purchases of computers (KSh 40,000 each) and LAN nodes are logged manually.

- **Monthly Payments**: Bandwidth subscriptions and fines are calculated manually.
- **Reporting**: Lists of institutions, defaulters, and disconnections are compiled manually.

2.2.1 System Structure

The existing manual system for managing Azani's internet service operations consists of several components and processes that work together to record, store, and manage institution-related information. Below is a detailed breakdown:

1. Institution Registration

Process: When an institution requests services, its details (name, type of institution, location, and contact person) are recorded manually in a registration book or spreadsheet. The registration fee of **KSh 8,500** is collected and recorded separately in a financial ledger.

Components:

- ✓ Registration book or spreadsheet for institution details.
- ✓ Financial ledger for recording registration fees.

2. Service Tracking

Process: Changes in services, such as bandwidth upgrades or infrastructure purchases (computers and LAN nodes), are updated manually in the records. Discounts for upgrades and fines for late payments are calculated by hand.

Components:

- Updated registration book or spreadsheet.
- Separate records for bandwidth changes and infrastructure purchases.

3. Infrastructure Management

Process: Details of infrastructure requirements, such as the number of computers or LAN nodes purchased, are recorded manually. Costs are calculated separately depending on the number of nodes or computers.

Components:

- Infrastructure logbooks or spreadsheets.
- Manual cost calculation sheets.

4. Financial Management

Process: Registration fees, installation fees, monthly subscriptions, fines, and reconnection charges are recorded in financial ledgers. Calculations for overdue fines (15%) and reconnection fees (KSh 1,000) are done manually.

Components:

- Financial ledgers for income and expenses.
- Manual calculations for fines, discounts, and reconnections.

5. Reporting

Process: Reports on registered institutions, defaulters, disconnections, and infrastructure requirements are prepared manually by compiling data from various sources. This process is tedious and often results in delayed or inaccurate reports.

Components:

- Compiled data from registration books, infrastructure logs, and financial ledgers.
- Manually prepared reports.

6. Field Assessment and Communication

Process: Azani staff visit institutions to assess readiness for installation (computers, LAN setup, number of users). Communication between staff and management is done manually through memos, phone calls, or physical reports.

Components:

- Field assessment forms.
- Manual communication channels (meetings, memos, phone calls).

Workflow of the Existing System:

- Institution registers and details are recorded manually.
- Registration fee is collected and logged in the financial ledger.
- Site visit is conducted to assess readiness.
- Installation fee is collected if requirements are met.
- Infrastructure purchases (computers, LAN nodes) are recorded separately.
- Monthly payments are tracked manually, with fines and reconnections calculated by hand.
- Reports are compiled manually from different records.
- Limitations of the System Structure

Fragmented Data Storage: Data is stored in separate books, spreadsheets, and ledgers, making it difficult to access and analyze information efficiently.

Manual Processes: Every step, from recording data to generating reports, is done manually, leading to inefficiencies and errors.

Lack of Integration: There is no centralized system to integrate and manage all institution-related data, resulting in duplication of efforts and inconsistencies.

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Limitation of the Existing System

1. Time-Consuming and Error-Prone

Manual Data Entry: Every institution's registration, payment, and infrastructure details are written down or entered into spreadsheets by hand. This takes a lot of time and increases the chance of mistakes such as miscalculations or missing entries.

Delayed Updates: When institutions upgrade bandwidth or purchase new infrastructure, staff must manually adjust records. This slows down service delivery and can cause inconsistencies.

Human Errors: Typing mistakes, misfiled records, or incorrect calculations (e.g., fines or discounts) are common in manual systems, leading to disputes and inefficiencies.

2. Fragmented Data Storage

Separate Files: Institution details, payment records, and infrastructure requirements are stored in different books, spreadsheets, or ledgers. This makes it difficult to get a complete picture of an institution's status.

Duplication of Effort: Staff often have to enter the same information in multiple places (e.g., registration book, payment ledger, and infrastructure log), which wastes time and increases the risk of inconsistencies.

Poor Integration: Since data is scattered, it's hard to link payments with service usage or infrastructure needs, making management decisions less accurate.

3. Inefficient Reporting

Manual Compilation: Reports on registered institutions, defaulters, disconnections, and infrastructure requirements are prepared by

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manually gathering data from different sources. This is slow and tedious.

Delayed Decision-Making: Because reports take time to prepare, management often works with outdated information, which affects planning and responsiveness.

Inaccurate Summaries: Errors in manual calculations or missing records can lead to unreliable reports, reducing trust in the system.

4. Poor Scalability as More Institutions Join

Growing Workload: As Azani expands and more institutions subscribe, the manual system becomes harder to manage. Each new institution adds more paperwork, calculations, and records to track.

Resource Strain: Staff must spend more time updating records, which increases operational costs and reduces efficiency.

Risk of Collapse: With hundreds of institutions, the manual system could become unmanageable, leading to service delays, billing errors, and dissatisfied clients.

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2.3 Overview of the Proposed System

2.3.1 System Overview

The **Azani Internet Service Provider Information System** will be a computerized database solution designed to automate registration, payment tracking, infrastructure management, and reporting. It will replace manual processes with an integrated, accurate, and user-friendly platform.

Key functionalities include:

- ✓ Institution registration and contact management.

- ✓ Payment tracking (registration, installation, monthly fees, fines, reconnections).
- ✓ Infrastructure management (computers and LAN nodes).
- ✓ Bandwidth upgrades with discount calculations.
- ✓ Automated reporting on institutions, defaulters, disconnections, and infrastructure needs.

2.3.2 Objectives of the Proposed System

General Objective

To provide an automated, efficient, and reliable solution for managing Azani's internet services and institutional records.

Specific Objectives

- ✓ **Automate Institution Registration and Contact Management**

The system will capture institution details (name, type, location) and contact person information in a centralized database.

This eliminates manual paperwork and ensures quick retrieval of records.

- ✓ **Record and Track All Payments**

Payments for registration, installation, monthly subscriptions, fines, and reconnections will be logged automatically.

The system will generate receipts and maintain transaction histories for accountability.

- ✓ **Manage Infrastructure Requirements and Sales**

Institutions purchasing computers or LAN nodes will have their orders recorded in the system.

This ensures accurate tracking of infrastructure costs and prevents loss of sales records.

- ✓ **Generate Automated Reports for Management**

Reports on registered institutions, defaulters, disconnections, and infrastructure requirements will be generated instantly.

This supports faster decision-making and reduces delays in administrative processes.

- ✓ **Compute Costs for Installations, Upgrades, Fines, and Reconnections**

The system will automatically calculate installation fees, discounts for bandwidth upgrades, overdue fines, and reconnection charges.

This reduces errors in financial management and ensures consistency.

- ✓ **Enhance Accountability and Transparency in Service Delivery**

By keeping accurate records of payments, infrastructure, and service status, the system ensures transparency.

Institutions can trust Azani's billing and service processes, strengthening customer relationships.

2.3.3 Benefits of the Proposed System

- ✓ **Efficiency**

Automates repetitive tasks such as registration, payment tracking, and reporting.

Saves time for staff and reduces workload.

- ✓ **Accuracy**

Eliminates human errors in calculations of fees, fines, and discounts.

Ensures financial records are consistent and reliable.

- ✓ **Transparency**

Provides clear records of transactions and service status.

Institutions can easily verify their payments and services, reducing disputes.

- ✓ **Reporting**

Generates timely and detailed reports on institutions, payments, and infrastructure.

Helps management make informed decisions quickly.

- ✓ **Scalability**

The system can handle increasing numbers of institutions without performance issues.

Supports Azani's growth and expansion into more schools and colleges.

- ✓ **Data Security**

Restricts access to authorized users only.

Protects sensitive financial and institutional data from unauthorized access or tampering.

2.3.4 Disadvantages of the Proposed System

- ✓ **Initial Cost of Implementation**

Purchasing computers, servers, and training staff requires upfront investment.

Smaller institutions may find this cost challenging initially.

- ✓ **Dependency on Electricity and Computers**

The system cannot function without power or hardware.

Power outages or hardware failures may temporarily disrupt operations.

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- ✓ **Possible Resistance from Staff**

Staff accustomed to manual processes may resist adopting the new system.

Training and change management will be necessary to ensure smooth transition.

- ✓ **Risks of Data Loss if Backups Are Not Maintained**

Without proper backup strategies, technical failures could result in loss of important records.

- ✓ **Regular backups and recovery plans are essential.**

Need for Periodic System Maintenance and Updates

As technology evolves, the system will require updates to remain secure and efficient.

Maintenance costs and technical support may be needed over time.

2.3.5 Scope of the System

In-Scope

- Institution registration and contact management.
- Payment tracking (registration, installation, monthly fees, fines, reconnections).
- Infrastructure management (computers and LAN nodes).
- Bandwidth upgrades with discount calculations.
- Automated reporting.

Out of Scope

- Online payment integration.
- Mobile application access.

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- External stakeholder access.

2.3.6 Cost and Benefits Analysis

Costs

- ❖ Development and testing (minimal if done in-house).
- ❖ Training staff and preparing manuals.
- ❖ Hardware purchases if required.
- ❖ Backup and maintenance costs.

Benefits

- ❖ Reduced administrative workload.
- ❖ Improved financial transparency.
- ❖ Accurate and timely reporting.
- ❖ Better planning and decision-making.

2.4 Feasibility Study

Technical Feasibility

The system will be developed using widely available database tools (e.g., Microsoft Access, MySQL, or PostgreSQL).

Azani already has functional computers and basic IT infrastructure.

Data storage requirements are moderate, focusing on institution records, payments, and infrastructure details.

Backup solutions (external drives or cloud storage) can be easily integrated. **Conclusion**: Technically feasible with existing resources.

Economic Feasibility

Development costs are minimal if built in-house.

Training and hardware purchases are one-time investments.

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Long-term savings from reduced paperwork, fewer errors, and faster reporting outweigh initial costs.

Improved financial transparency reduces risks of revenue loss. **Conclusion**: Economically feasible with high return on investment.

Operational Feasibility

The system simplifies workflows, reducing manual errors.

Staff and administrators will find it easy to use due to a simple interface.

Training requirements are minimal.

Automated reporting enhances decision-making. **Conclusion**: Operationally feasible and user-friendly.

Legal Feasibility

The system complies with Kenyan data protection regulations.

Sensitive institutional and financial records will be restricted to authorized users.

Licensed software will be used, avoiding copyright issues. **Conclusion**: Legally feasible and compliant.

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Schedule Feasibility

Development and testing can be completed within a few weeks.

Implementation can align with the start of a new billing cycle or academic term.

The system can be scaled gradually to accommodate more institutions. **Conclusion**: Schedule feasible within project timelines.

Overall Feasibility Conclusion

The proposed Azani Internet Service Provider Information System is **technically, economically, operationally, legally, and schedule feasible**. It will enhance efficiency, accuracy, transparency, and scalability in Azani's operations, making it a sustainable solution for managing internet services across educational institutions.

2.5 Fact-Finding

Fact-finding is the process of gathering relevant information about the **Azani Internet Services Provider Information System** to understand its requirements, user needs, and challenges in the existing manual system. This step ensures that the proposed system is designed based on accurate data and tailored to the operational realities of Azani.

Objectives of Fact-Finding

The main objectives of fact-finding for this project are:

- To understand the current manual system used to manage institution registration, payments, and service delivery.
- To identify the challenges faced by institutions, Azani staff, and administrators in managing internet services and infrastructure.
- To collect user requirements and determine the expectations of the new system.
- To analyze financial transactions related to registration, installation, monthly subscriptions, fines, and reconnections.
- To evaluate the feasibility of automating Azani's operations and improving efficiency.

Fact-Finding Methods Used

To gather information for system development, the following fact-finding methods were used:

Observation

The current manual system of managing institutions was observed to understand how registrations, payments, and infrastructure records are handled.

Challenges noted included delays, data duplication, and errors in financial tracking.

Interviews

Interviews were conducted with Azani staff, administrators, and selected institution representatives to gather insights on the existing system.

The interviews helped identify common problems such as late payment tracking, difficulty in generating reports, and inefficiencies in service upgrades.

Questionnaires

Structured questionnaires were distributed to institutions to collect feedback on registration, payment processes, and service delivery.

Responses highlighted the need for a user-friendly system that simplifies billing, infrastructure management, and reporting.

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Document Analysis

Existing manual records, such as registration books, payment ledgers, and infrastructure logs, were reviewed.

This helped in understanding data storage needs, report generation requirements, and financial management processes.

Summary of Findings

Based on the fact-finding process, the following key findings were established:

- The current system is manual, leading to inefficiencies in managing registrations, payments, and infrastructure records.
- Data retrieval is slow and prone to errors, making it difficult for staff to track institutions and financial transactions.
- There is no centralized database, which leads to duplication of records and loss of important information.
- Financial reports are difficult to generate manually, making it challenging to track subscriptions, fines, and reconnections.
- Institutions and Azani staff require a user-friendly system that simplifies registration, financial management, and reporting.
- The fact-finding process provided valuable insights into the problems of the existing manual system and the requirements for the new system. The findings will be used to design a database system that improves efficiency, accuracy, and accessibility in managing Azani's internet services.

2.6 Summary of Fact-Finding Methods Used

The fact-finding process was essential in gathering relevant information about Azani's current service management system and understanding the requirements for the proposed system. Several methods were employed to collect data, each contributing valuable insights into the system's needs and user expectations.

Observation

Purpose: To understand the workflow and challenges of the current manual registration, payment, and service tracking system.

Outcome: Revealed inefficiencies such as delays in updating records, errors in financial tracking, and lack of centralized data.

Interviews

Purpose: To gather insights from key stakeholders such as Azani staff, administrators, and institution representatives.

Outcome: Highlighted the need for an automated system to streamline service management, reduce errors, and improve data accessibility.

Questionnaires

Purpose: To collect structured feedback from a larger group of institutions.

Outcome: Confirmed findings from interviews and observations, showing a desire for a more efficient and user-friendly system to manage services and payments.

Document Analysis

Purpose: To analyze existing records and documentation related to institution registrations, payments, and infrastructure.

Outcome: Revealed duplication of records and inefficiency in generating accurate financial reports.

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2.7 System Requirements and Specifications

The minimum hardware and software requirements for the Azani Internet Services Provider Information System to run effectively and efficiently are tabulated as follows:

Software Requirements

1) Computer running Microsoft Windows 7 (Professional) or above
2) Microsoft Access application program
3) Microsoft Word application program
4) Kaspersky Antivirus Program (Latest Version)

Hardware Requirements

1) 45 GB of Hard Drive space – to store all the data entered
2) 5 GB RAM
3) 3.06 GHz Dual Core Processor
4) 20” LED monitor
5) 2 LaserJet ink Printers for report hardcopy output
6) Memory stick or CD/DVD drive to store backups of data

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CHAPTER THREE : SYSTEM DESIGN

3.1 Flowcharts Design

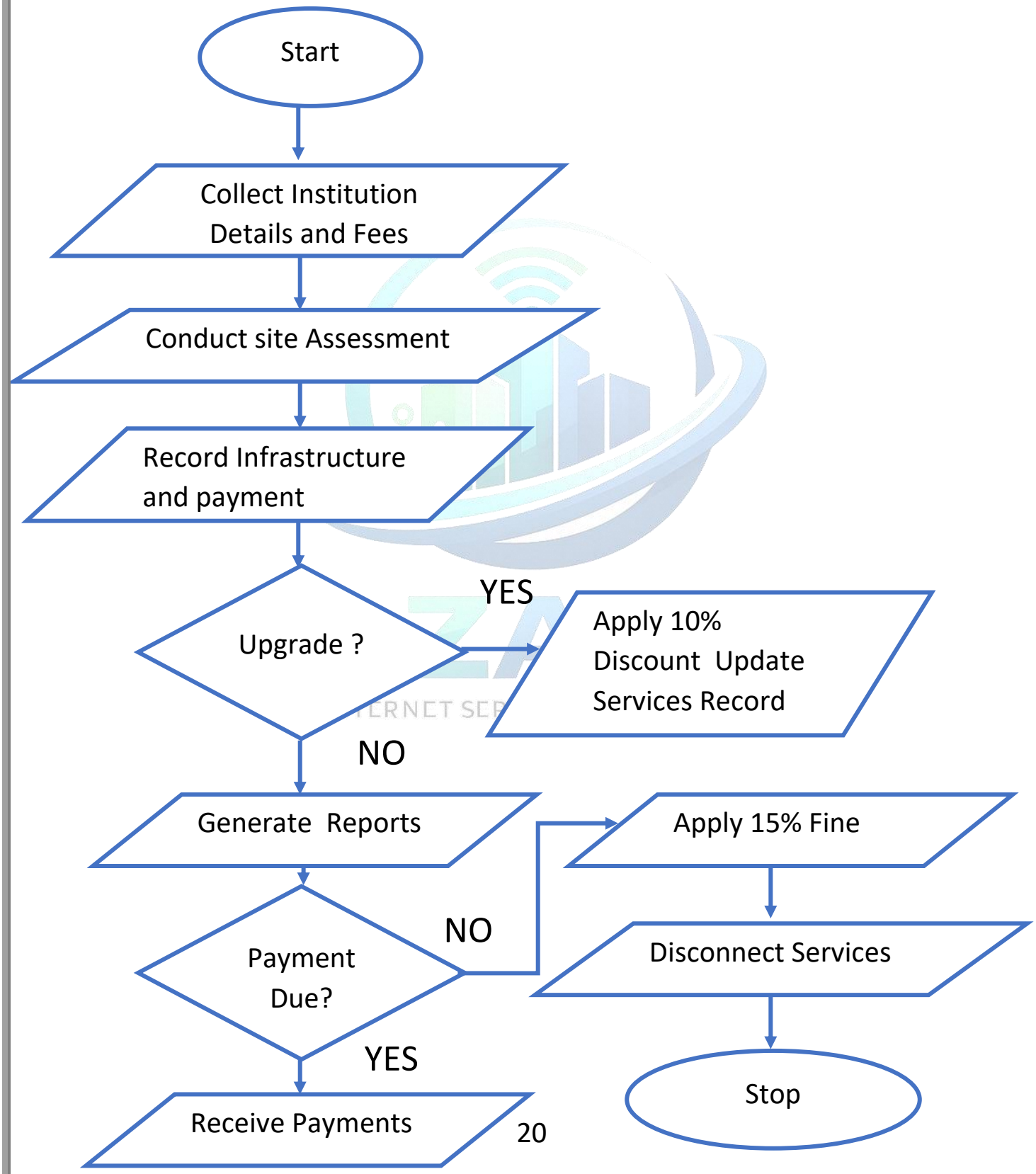

Figure 1: Flowchart1

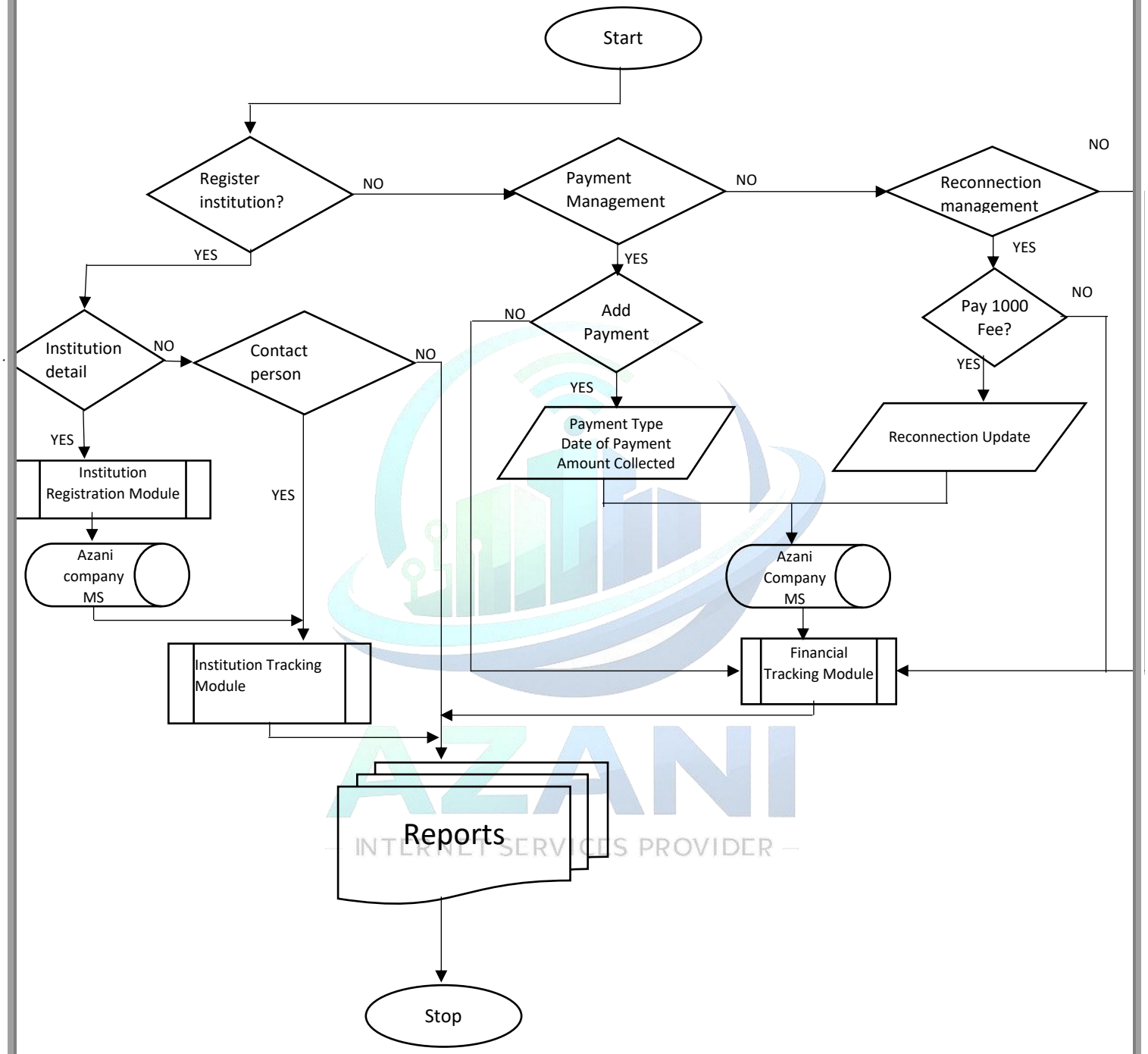

Figure 2:Flowchart2

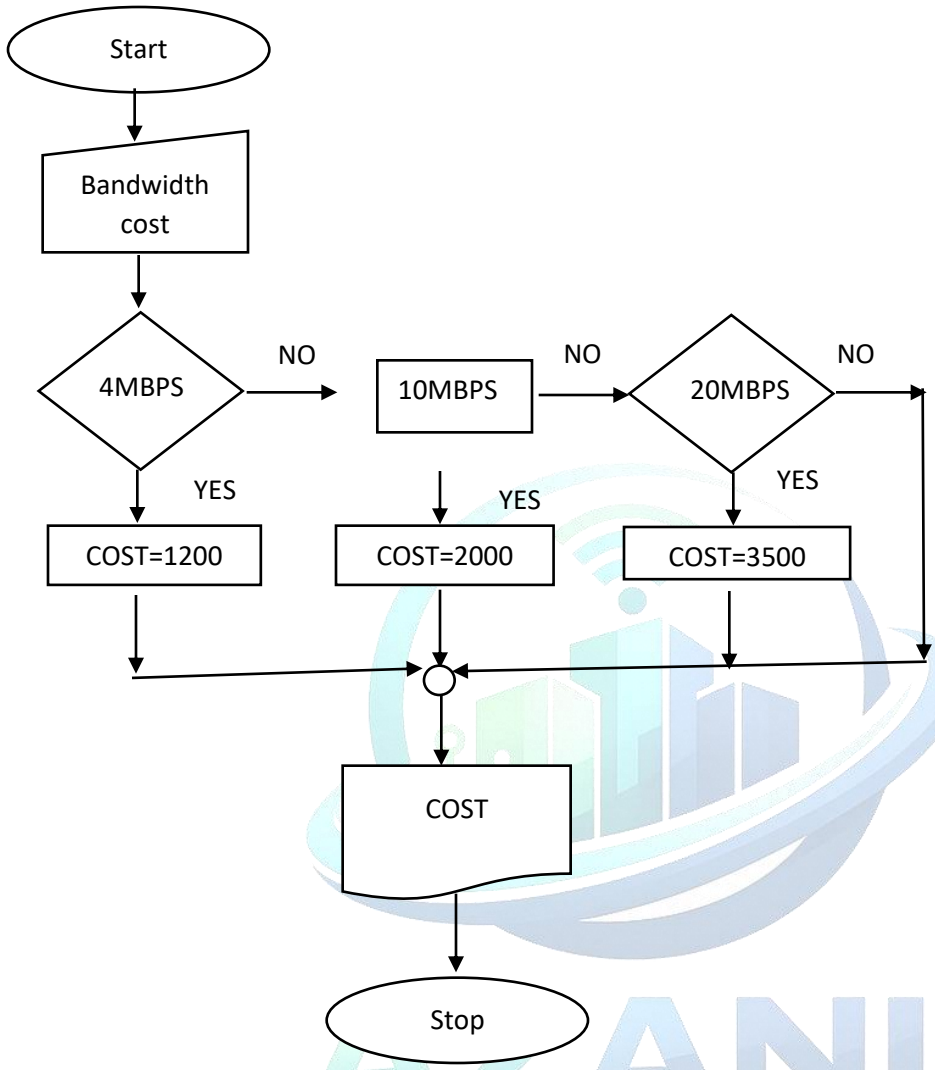

Figure 3: Flowchart3

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3.2 Table Design

1. Institutions Table

Stores details of registered institutions.

Field Name	Data Type	Description
InstitutionID	AutoNumber	Primary key
InstitutionName	Text	Name of the institution
InstitutionType	Text	Primary, Junior, Senior, College
Location	Text	Physical address
ContactPerson	Text	Name of contact person
PhoneNumber	Text	Contact phone number
Email	Text	Contact email
RegistrationDate	Date/Time	Date of registration

2. Payments Table

Tracks all financial transactions.

Field Name	Data Type	Description
PaymentID	AutoNumber	Primary key
InstitutionID	Number	Foreign key from Institutions
PaymentType	Text	Registration, Installation, Monthly
Amount	Currency	Amount paid
PaymentDate	Date/Time	Date of payment
Status	Text	Paid, Overdue, Reconnected

3. Infrastructure Table

Records infrastructure purchases.

Field Name	Data Type	Description
InfrastructureID	AutoNumber	Primary key
InstitutionID	Number	Foreign key from Institutions
ItemType	Text	Computer or LAN Node
Quantity	Number	Number of items
UnitCost	Currency	Cost per item
TotalCost	Currency	Computed field (Quantity × UnitCost)

4. Bandwidth Table

Stores bandwidth subscription details.

Field Name	Data Type	Description
BandwidthID	AutoNumber	Primary key
InstitutionID	Number	Foreign key from Institutions
BandwidthLevel	Text	4MBPS, 10MBPS
MonthlyCost	Currency	Cost per month
UpgradeStatus	Text	Upgraded, Standard
DiscountApplied	Yes/No	Whether 10% discount was applied

5. Reports Table

Stores generated reports and summaries.

Field Name	Data Type	Description
ReportID	AutoNumber	Primary key
InstitutionID	Number	Foreign key from Institutions
ReportType	Text	Defaulters, Disconnections, Summary
GeneratedDate	Date/Time	Date of report generation
Notes	Memo	Additional comments or findings

6. Service Status Table

Tracks service connection status.

Field Name	Data Type	Description
StatusID	AutoNumber	Primary key
InstitutionID	Number	Foreign key from Institutions
ConnectionStatus	Text	Active, Disconnected, Pending
LastPaymentDate	Date/Time	Most recent payment
FineApplied	Currency	15% overdue fine
ReconnectionFee	Currency	KSh 1,000 if applicable

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3.3 Input Design

1. Register Institution form

Register Institution

Institution Details

Institution Name: Primary

Institution Type:

Location: :

Contact Person

Contact Name:

Phone Number:

Email Address:

Billing Details

Band Width Plan:

Infrastructure

Computers

LAN Nodes:

KSH 10,000

Save Institution | clear | Delete Institution

2. Payment Form

Institution Details

Institution Name:

Contact person:

Service Status: Active

Billing Details

Band-Width Plan 4MPS

Monthly fee: Ksh1200

Overdue Fine(%15): 0.0

Reconnection Fee:

TOTAL COST

Payment Info

Payment Date:

Payment Amount:

Email Address:

Payment Method : Cash

Mobile

Payment history

Submit Payment | clear | Print Invoice

3.Reconnection Management Form

Institution Details

Institution Name:

Contact person:

Service Status: Disconnected

Billing Details

Band-Width Plan 4MPS

Monthly fee: Ksh1200

Overdue Fine(%15): 100

Reconnection Fee: 1000

TOTAL COST

Payment Info

Payment Date:

Payment Amount:

Email Address:

Payment Method : ☐ Cash ☐ Mobile

Payment history

Submit Payment | Process Reconnection | Apply overdue

3.4 OUTPUT DESIGN

Registered institution report

This output screen display a list of all the registered institutions on the company.

Registered Institutions

InstitutionID	Institution Name	Institution Type	Date Registered	Location
		Total Institution		

Defaulters Report

This screen display the list of all the defaulters institution and the overdue fines they should pay.

Defaulters Report

InstitutionID	Institution Name	Remaining days	Due Amount	Status
		Total Institution		

Disconnected Institution Report

This output screen display the list of all the disconnected institution that have registered and the total amount they should pay for their reconnection back to the server .

Disconnection Report

InstitutionID	Institution Name	Reconnection fee	Date disconnect	Status
		Total Institution		

CHAPTER FOUR: SYSTEM CONSTRUCTION

4.0 Introduction

The System Construction phase marks a significant milestone in the development lifecycle of the proposed computerized database system for **Azani Company**. This phase involves the actual implementation of the system design, transforming conceptual ideas and requirements into a functional and operational solution. Throughout this chapter, we will delve into the various aspects of system construction, including database creation, application development, integration of functionalities, and testing procedures.

In this chapter, we will explore the following key areas:

1. **Database Creation**: We will outline the process of creating the database schema, defining tables, relationships, and constraints based on the specified system requirements and design specifications. Emphasis will be placed on data integrity, normalization, and optimization to ensure efficient data storage and retrieval.
2. **Application Development**: We will discuss the development of the application layer, encompassing user interfaces, business logic, and application functionalities. Attention will be given to usability, functionality, and adherence to design principles to deliver a user-friendly and intuitive system.
3. **Integration of Functionalities**: We will examine the integration of various system functionalities, including institution registration, service management, billing, reporting, and security mechanisms. The goal is to ensure seamless interaction and interoperability between different system components to deliver a cohesive and integrated solution.

4. **Testing Procedures**: We will explore the testing methodologies and procedures employed to validate the functionality, performance, and reliability of the system. This includes unit testing, integration testing, system testing, and user acceptance testing to identify and rectify any issues or defects before deployment.

Throughout the System Construction phase, collaboration between developers, designers, stakeholders, and end-users is crucial to ensure that the final product aligns with business objectives, meets user needs, and adheres to quality standards. By following established best practices and methodologies, we aim to deliver a robust, scalable, and reliable computerized database system that empowers **Azani Company** to optimize its internet service operations and achieve its strategic goals.

4.1 DATABASE RELATIONSHIP DIAGRAM

The database relationship diagram is an implementation of the design shown earlier.

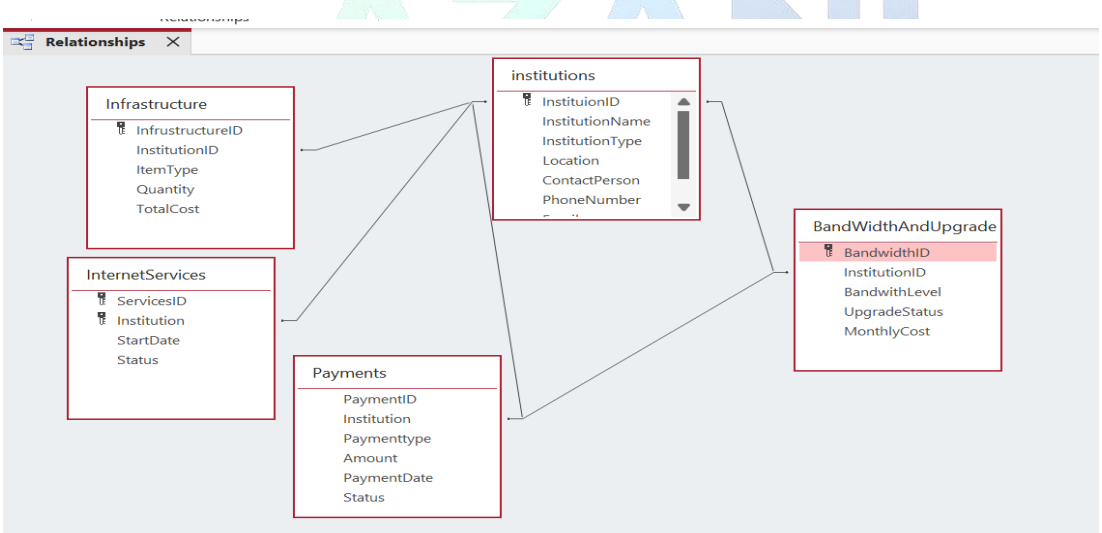

Figure 4: Database Relationship

4.2 ENFORCING INTEGRITY CONSTRAINTS

In the System Construction phase, it is crucial to enforce integrity constraints within the database to maintain data accuracy, consistency, and reliability. Integrity constraints are rules that govern the validity and accuracy of data stored in the database, ensuring that it remains consistent and meaningful. The following integrity constraints will be enforced in the proposed computerized database system for **Azani Company**:

1. **Entity Integrity Constraint** Each table will have a primary key constraint, ensuring that each record within the table is uniquely identifiable. Primary keys will be defined for tables such as *Institutions*, *InternetServices*, *Installations*, *Payments*, and *LANNodes* to uniquely identify records within these entities.
2. **Referential Integrity Constraint** Foreign key constraints will be enforced to maintain referential integrity between related tables. For example, the *Payments* table will have a foreign key referencing the *Institutions* table, ensuring that each payment record is associated with a valid institution.
3. **Domain Integrity Constraint** Domain constraints will be applied to ensure that data values within each attribute conform to specified data types and formats. For instance, the *Telephone* attribute in the *Institutions* table may be constrained to accept only numeric values with a specific length.
4. **Check Constraints** Check constraints will be utilized to enforce specific conditions or rules on data values within attributes. For example, a check constraint may be applied to the *Cost* attribute in the *InternetServices* table to ensure that it is a positive numeric value.

5. **Unique Constraints** Unique constraints will be enforced to ensure that no duplicate values exist within specified attributes. Unique constraints may be applied to attributes such as institution names or email addresses to prevent duplication of records.
6. **Default Constraints** Default constraints will be defined to specify default values for attributes when no explicit value is provided during data entry. For example, a default constraint may be applied to the *DateRegistered* attribute in the *Institutions* table to automatically set the current date when a new institution record is created.

By enforcing these integrity constraints, the proposed database system will maintain data consistency, accuracy, and reliability, ensuring that it remains a trusted source of information for Azani Company's internet service operations. Violations of integrity constraints will be prevented at the database level, reducing the risk of data corruption and ensuring the overall quality of the system.

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4.3 DATABASE TABLES

4.3.1 INSTITUTIONS TABLE

The *Institutions* table stores core client information. **Key fields include:**

- **InstitutionID (PK):** Unique identifier for each institution.
- **InstitutionName:** Name of the school or college.
- **InstitutionType:** Specifies whether it is *Primary*, *Junior*, *Senior*, or *College*.
- **Location:** Physical address of the institution.
- **ContactPerson:** Name of the person responsible.
- **PhoneNumber:** Contact number.
- **Email:** Institution's email address.

This table is the backbone of the system, linking all other tables through relationships.

institutions

Field Name	Data Type	Description (Optional)
InstituionID	AutoNumber	Primarykey
InstitutionName	Short Text	Name of institutions
InstitutionType	Short Text	Primary,junior, senior, college
Location	Short Text	physical address
ContactPerson	Short Text	Name of contctperson
PhoneNumber	Short Text	Contact phonenumber
Email	Short Text	Contact email
RegistrationDate	Date/Time	Date of registration

Figure 5: institution table

4.3.2 INTERNETSERVICES TABLE

The *InternetServices* table stores details of the internet packages subscribed to by institutions. **Key fields include:**

- **ServicesID (PK):** Unique identifier for each service record.
- **InstitutionID (FK):** Links the service to a registered institution.
- **StartDate:** Date when the service was activated.
- **Status:** Indicates whether the service is *Active*, *Suspended*, or *Disconnected*.

This table ensures that each institution's internet subscription is tracked accurately.

Field Name	Data Type
ServicesID	AutoNumber
Institution	Number
StartDate	Date/Time
Status	Short Text

Figure 6: Internet services

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4.3.3 PAYMENTS TABLE

The *Payments* table records all financial transactions made by institutions. **Key fields include:**

- **PaymentID (PK):** Unique identifier for each payment.
- **InstitutionID (FK):** Links the payment to the institution making it.
- **PaymentType:** Specifies whether it is *Registration*, *Monthly Fee*, *Installation*, or *Reconnection*.
- **Amount:** The amount paid in KSh.
- **PaymentDate:** Date of payment.
- **Status:** Indicates whether the payment is *Completed*, *Pending*, or *Overdue*.

This table enforces billing rules, overdue fines, and reconnection charges.

Payments

Field Name	Data Type	Description (Optio
PaymentID	AutoNumber	
Institution	Number	
Paymenttype	Short Text	
Amount	Number	
PaymentDate	Date/Time	
Status	Short Text	Paid Overdue Disconnected

Figure 7: Payment Table

4.3.4 INFRASTRUCTURE TABLE

The *Infrastructure* table captures details of equipment and infrastructure purchased or installed. **Key fields include:**

- **InfrastructureID (PK):** Unique identifier for each infrastructure record.
- **InstitutionID (FK):** Links infrastructure to the institution.
- **ItemType:** Specifies whether the item is a *PC* or *LAN Node*.
- **Quantity:** Number of items purchased.
- **TotalCost:** Total cost of the infrastructure.

This table helps track readiness of institutions for internet installation.

Infrastructure

Field Name	Data Type
InfrustructureID	AutoNumber
InstitutionID	Number
ItemType	Short Text
Quantity	Number
TotalCost	Calculated

Figure 8: infrastructure Table

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4.3.5 BANDWIDTHANDUPGRADE TABLE

The *BandwidthAndUpgrade* table manages bandwidth subscriptions and upgrades. **Key fields include:**

- **BandwidthID (PK):** Unique identifier for each bandwidth record.
- **InstitutionID (FK):** Links bandwidth to the institution.
- **BandwidthLevel:** Specifies the subscribed bandwidth (e.g., 4 MBPS, 10 MBPS, etc.).
- **UpgradeStatus:** Indicates whether the institution has upgraded bandwidth.
- **MonthlyCost:** Monthly cost of the subscribed bandwidth.

This table enforces upgrade discounts and ensures accurate monthly billing.

Field Name	Data Type
BandwidthID	AutoNumber
InstitutionID	Number
BandwithLevel	Short Text
UpgradeStatus	Short Text
MonthlyCost	Calculated

Figure 9: Bandwidth Subscription

4.4 DATABASE QUERIES

In order to combine several tables, we need to create queries that would be used to process various transactions and generate important reports

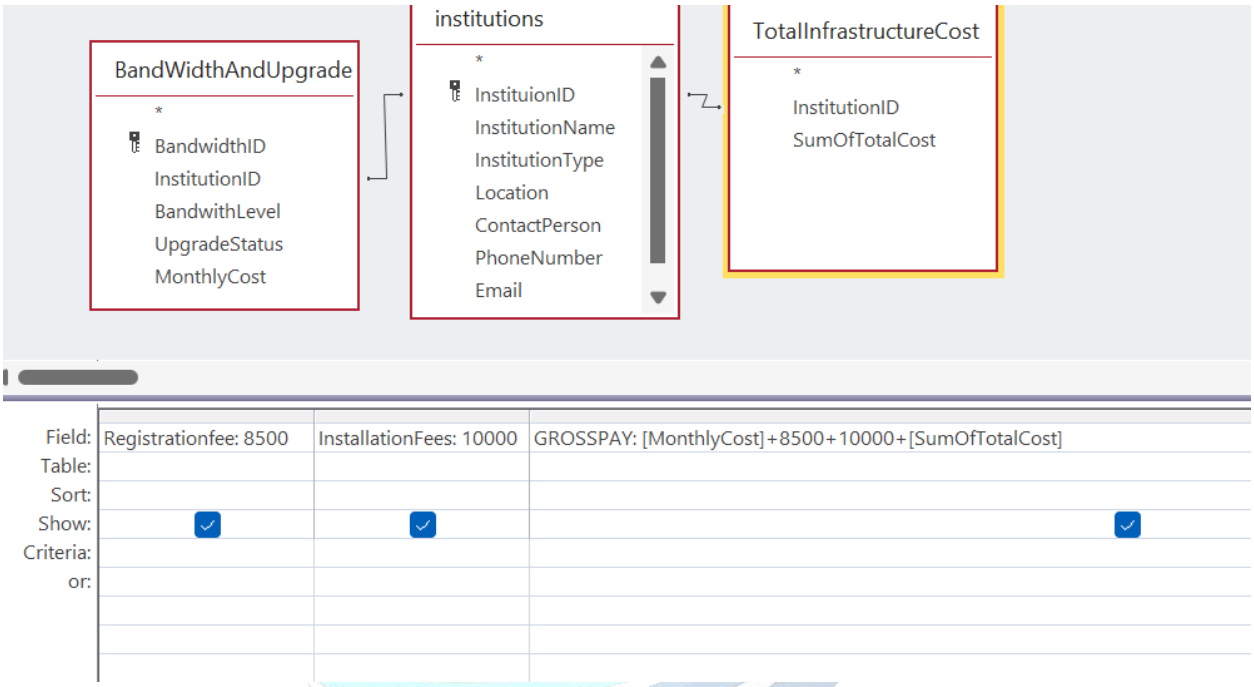

Figure 10: Total Installation Cost

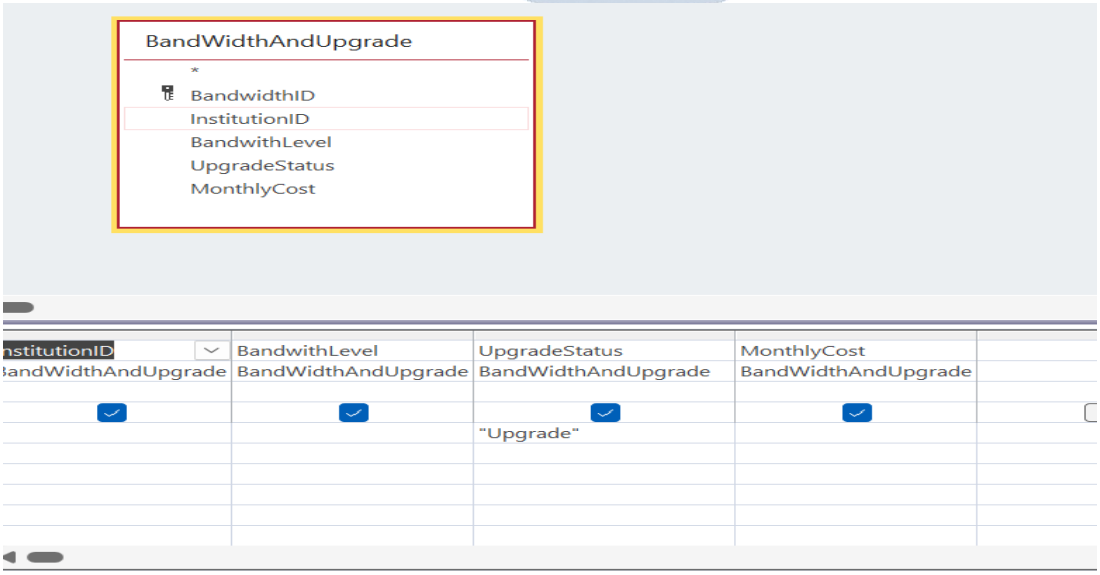

Figure 11: Upgrade query

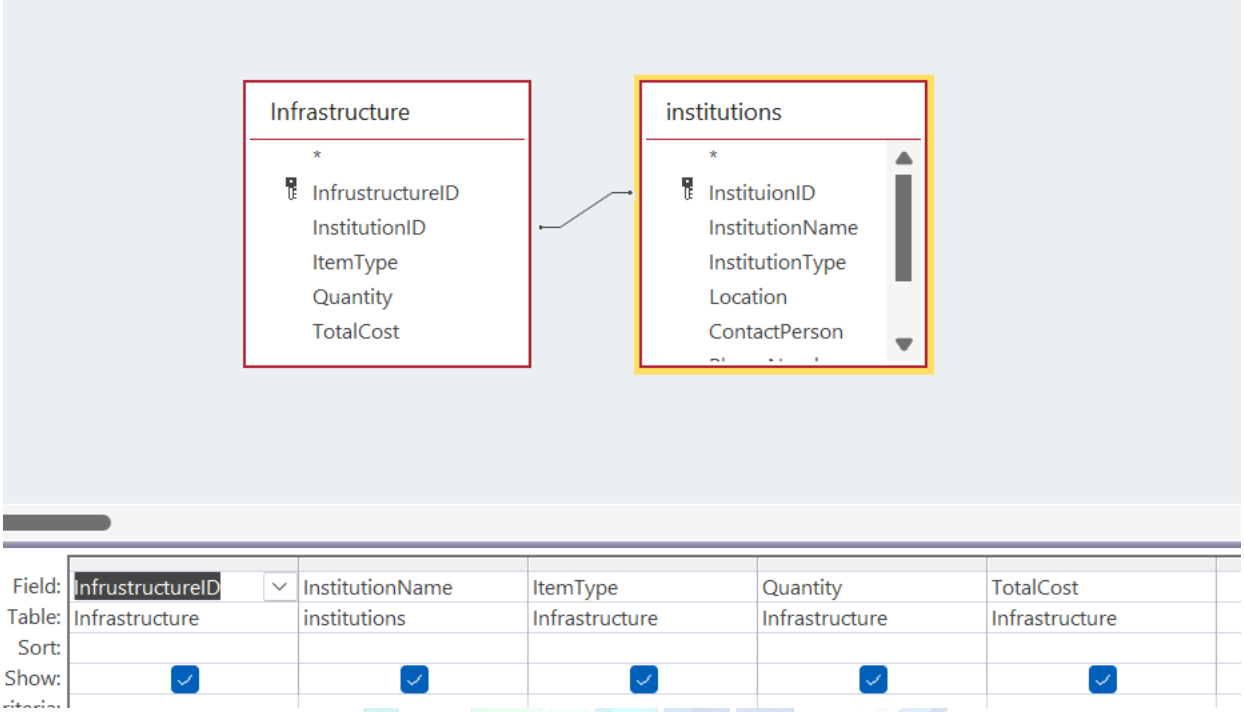

Figure 12: Infrastructure

4.5 DATA MANIPULATION

Data is processed in calculated field as follows

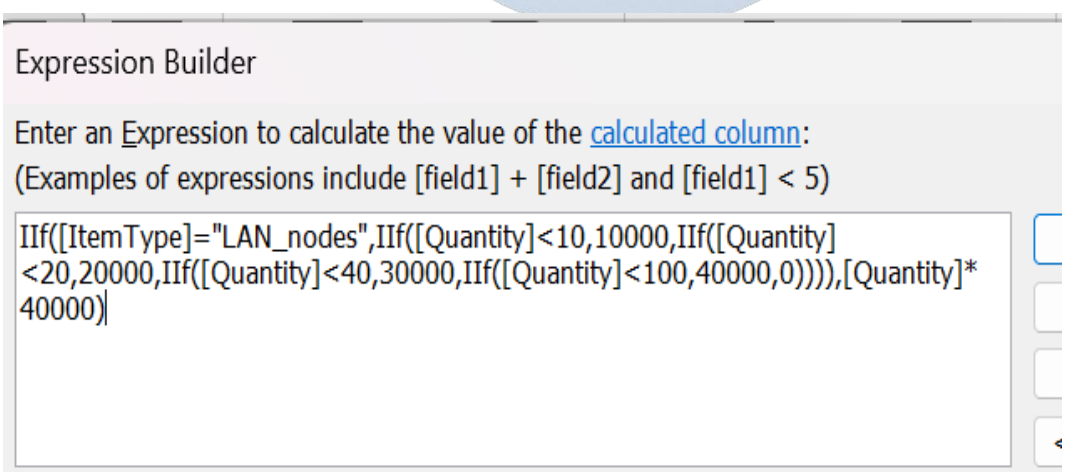

Figure 13: LAN MANIPULATION

Enter an Expression to calculate the value of the calculated column:

(Examples of expressions include [field1] + [field2] and [field1] < 5)

```
IIf([UpgradeStatus]="Upgrade",IIf([BandwithLevel]="4MBPS",1200*
0.9,IIf([BandwithLevel]="10MBPS",3500*0.9,IIf([BandwithLevel]="20MBPS",3500*
0.9,IIf([BandwithLevel]="25MBPS",4000*0.9,IIf([BandwithLevel]="50MBPS",7000*
0.9,0))))),IIf([BandwithLevel]="4MBPS",1200,IIf([BandwithLevel]="10MBPS",3500,
IIf([BandwithLevel]="20MBPS",3500,IIf([BandwithLevel]="25MBPS",4000,IIf([Band
withLevel]="50MBPS",7000,0))))))
```

Figure 14: BANDWIDTH

4.6 FORMS

In order to facilitate user interaction with the database, forms have been designed to provide a simple and intuitive interface for data entry, updating, and retrieval. Forms act as the front-end of the system, allowing users to interact with the underlying tables without directly accessing raw data.

The forms developed for the **Azani Internet Service Provider Information System** serve the following purposes:

- **Institution Registration Form:** Enables entry of institution details such as name, type, location, and contact information.
- **Service Subscription Form:** Allows users to select bandwidth packages, record installation details, and manage upgrades.
- **Payment Form:** Provides a structured interface for recording registration fees, monthly payments, fines, and reconnection charges.

- **Infrastructure Form:** Facilitates entry of purchased PCs and LAN nodes, ensuring accurate tracking of institution readiness.

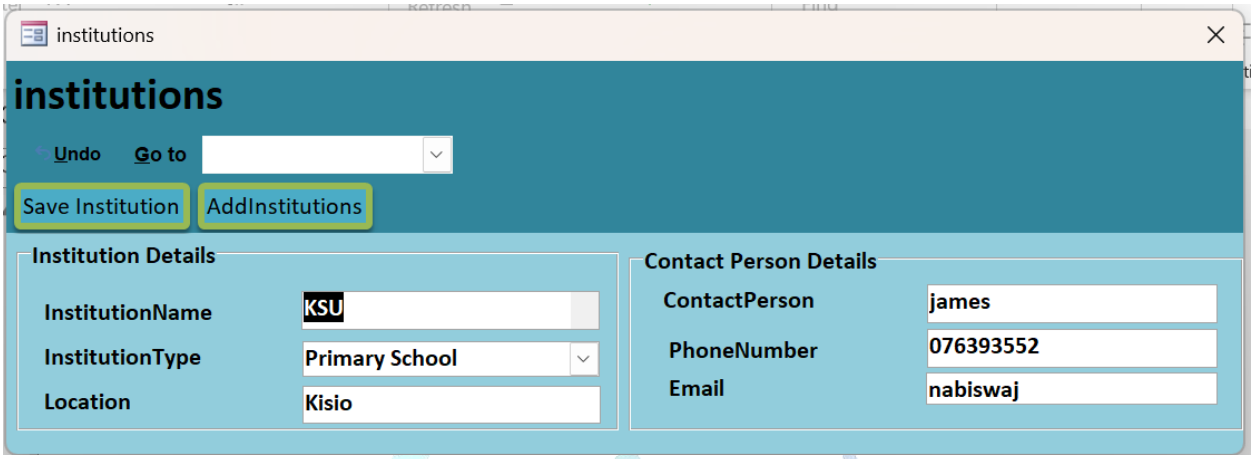

Figure 15: Institution registration form

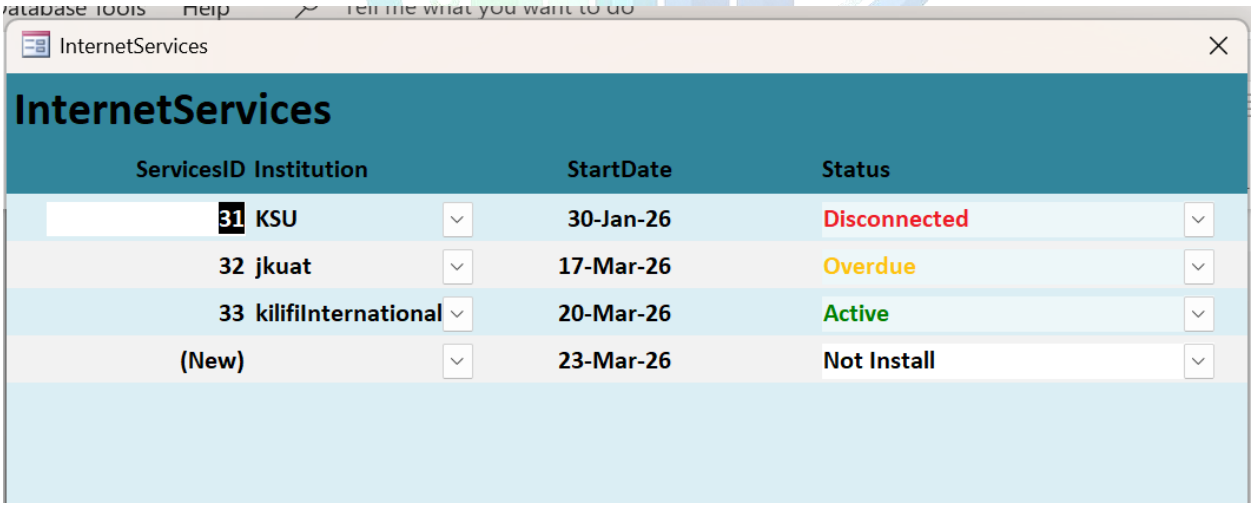

Figure 16: Manage subscription

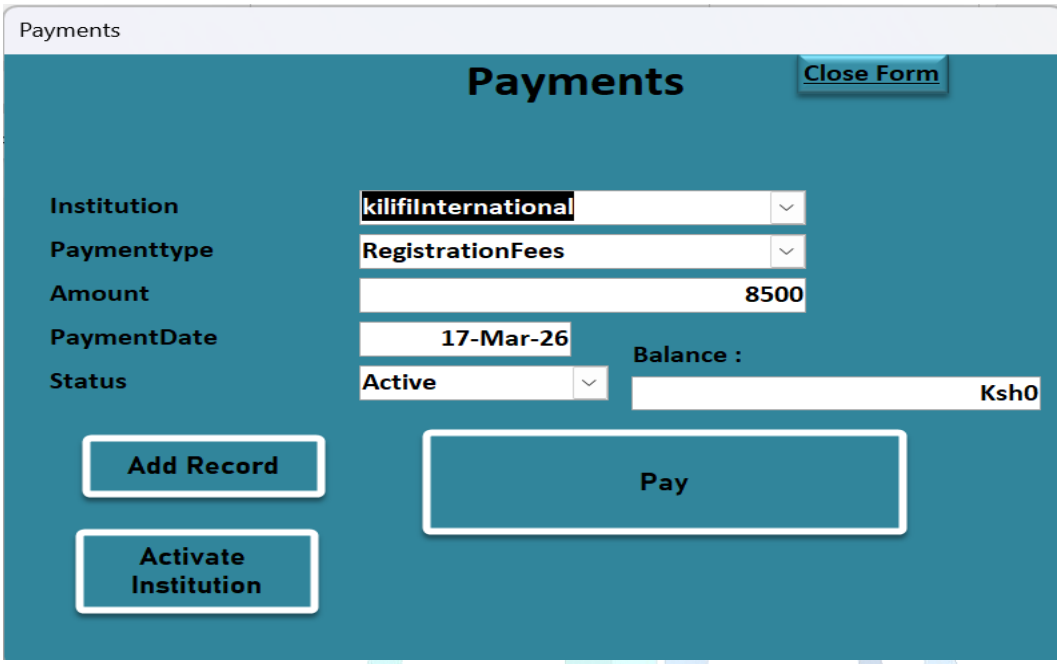

Figure 17: Payment form

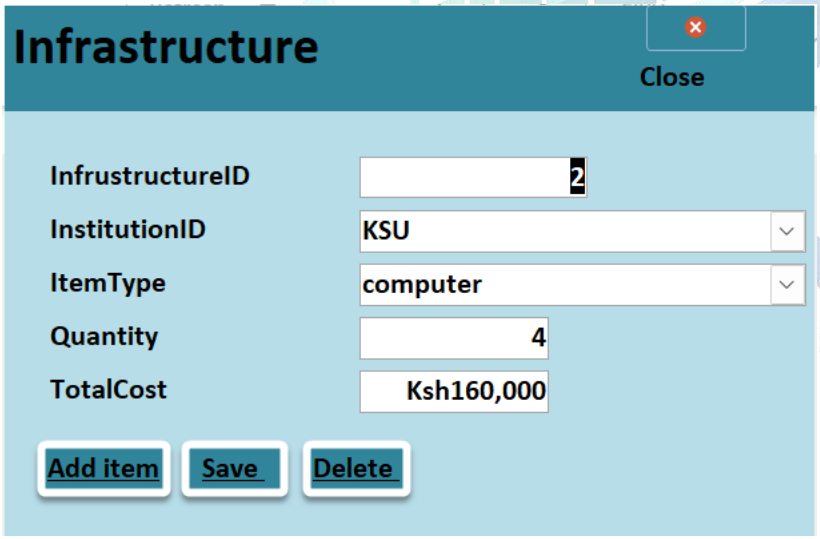

Figure 18: Infrastracture form

4.7 REPORT DESIGN

Reports are an essential component of the Azani Internet Service Provider Information System, providing summarized and structured information from the database for decision-making and operational monitoring. Unlike forms, which are primarily used for data entry and updates, reports are designed to present data in a clear, organized, and printable format.

Registered Institutions

InstitutionName	InstitutionType	Location	RegistrationDate
KitaleSchool	Primary School	kitale	11/03/2026
kilifiInternational	Senior School	Kilifi	18/03/2026
KSU	Primary School	Kisio	11/03/2026
jkuat	Junior School	Nakuru	20/03/2026

Figure 19: Registered Institution

Disconnection List

ServicesID	Institution	Status	MonthlyCost	OverdueFine	ReconnectionFees	TotalCharge
31	KSU	Disconnected	Ksh1,080	Ksh162	Ksh1,000	Ksh2,242

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Figure 20: Disconnection List

Defaulters List

ServicesID	Institution	Status	MonthlyCost	Fine	TotalCharge
32	jkuat	Overdue	Ksh3,600	540	4140

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Figure 21: Overdue list

UpgradeMonthlyPay

UpgradeMonthlyPay

InstitutionID	BandwithLevel	UpgradeStatus	MonthlyCost
KSU	4MBPS	Upgrade	Ksh1,080
jkuat	25MBPS	Upgrade	Ksh3,600

Figure 22: Monthly Upgrade

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CHAPTER FIVE : USER MANUAL

5.1 INTRODUCTION

The user manual provides comprehensive guidance and instructions for users to effectively utilize the proposed computerized database system for **Azani Company**. This manual serves as a reference document for users, outlining system functionalities, navigation, and procedures for performing various tasks within the system. Below is an overview of the contents typically included.

System Overview The Azani Internet Service Provider Information System is a comprehensive database application developed to streamline the management of internet services for learning institutions. The system facilitates the registration of institutions, recording of service subscriptions, calculation of monthly fees, management of infrastructure requirements, generation of reports, and more, all within a user-friendly and intuitive interface.

User Manual Contents This user manual is organized into sections that cover various aspects of the system's functionality, including institution registration, service management, payment processing, reporting, and more. Each section provides step-by-step instructions, screenshots, and tips to guide you through using the system effectively.

Whether you are a new user seeking basic instructions or an experienced user looking for advanced features, this manual is designed to accommodate users of all skill levels. We encourage you to explore the manual thoroughly and leverage its resources to maximize your productivity and proficiency with the Azani Internet Service Provider Information System.

5.1.1 SOFTWARE REQUIREMENTS

To use the Azani Internet Service Provider Information System, you will need the following software:

- **Microsoft Access:** The system is built using Microsoft Access as the primary database management software. Ensure that you have access to a licensed copy of Microsoft Access to run the application.
- **Web Browser:** For accessing web-based interfaces or reports generated by the system, a standard web browser such as Chrome, Firefox, or Edge is recommended.

5.1.2 HARDWARE REQUIREMENTS

To ensure optimal performance of the Azani Internet Service Provider Information System, your computer hardware should meet the following minimum requirements:

- **Operating System:** The system is compatible with Windows operating systems, including Windows 7, 8, or 10.
- **Processor:** A minimum of Intel Core i3 or equivalent processor is recommended for smooth operation.
- **Memory (RAM):** At least 4GB of RAM is recommended to handle database operations and multitasking efficiently.
- **Storage:** Sufficient disk space is required to store the system application files and database. A minimum of 500GB of available storage is recommended.

5.2 HOW TO INSTALL, START AND RUN THE SYSTEM INTERFACE

To launch the system:

1. Click on the **Azani ISP System icon** to open the application.

2. On first-time use, a screen appears prompting the user to **Enable Content**.
3. Click on **Enable Content** to activate the system and begin using the interface.

Figure 23: Loading Interface

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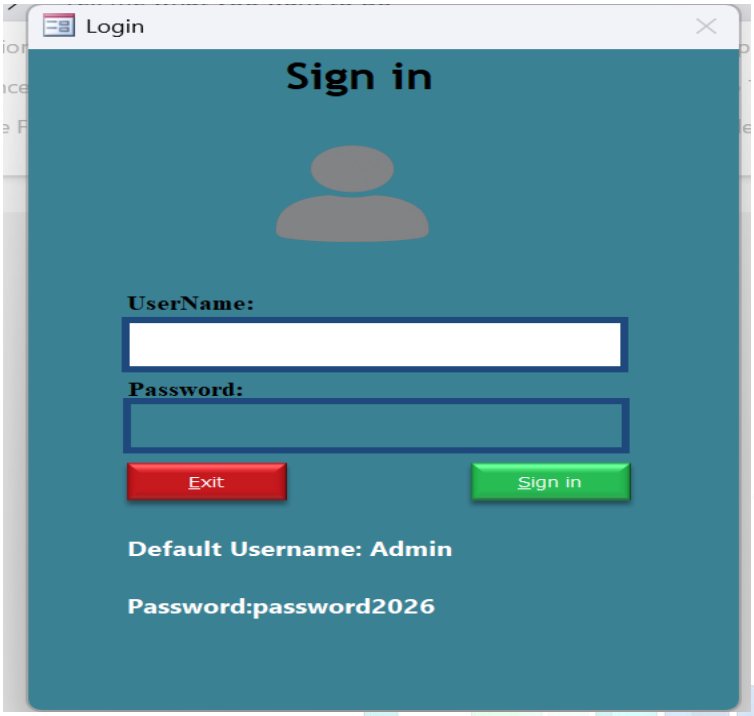

Figure 24: Login

The Dashboard will display as shown

Figure 25: Dashboard

CHAPTER 6: SYSTEM IMPLEMENTATION

6.1 STAFF TRAINING

Staff training is a critical component of the system implementation process, ensuring that employees have the necessary knowledge and skills to effectively use the computerized database system for **Azani Company**. The training program aims to familiarize staff with the system's functionalities, features, and workflows, enabling them to perform their job responsibilities efficiently and confidently.

Below are key aspects included in the staff training documentation:

1. **Training Objectives** The objectives of the training program are to ensure that staff members understand how to navigate the system, manage institution records, process payments, and generate reports. Emphasis is placed on system proficiency as a means of enhancing productivity, reducing errors, and improving service delivery to institutions.
2. **Training Topics** Training sessions will cover:
    - System navigation and menu structures.
    - Institution registration and client management.
    - Service subscription and bandwidth upgrades.
    - Payment recording, overdue fines, and reconnection charges.
    - Infrastructure management (PCs and LAN nodes).
    - Report generation and analysis.
    - Data security and user access controls.

3. **Training Methods** A blended approach will be used, including instructor-led sessions, demonstrations, hands-on exercises, and e-learning modules. This ensures that different learning styles are accommodated. Staff will be encouraged to practice using the system in a controlled environment before handling live data.
4. **Training Schedule** Training will be conducted in phases, with sessions scheduled for different departments such as billing, installations, and customer service. Each session will last between 2–3 hours, with refresher courses offered periodically.
5. **Training Materials** Comprehensive training materials will be developed, including user manuals, quick reference guides, video tutorials, and FAQs. These resources will remain accessible for staff to consult when needed.
6. **Trainer Qualifications** Trainers will be selected based on their expertise in Microsoft Access and their understanding of Azani's business processes. They must also demonstrate effective communication and training delivery skills.
7. **Hands-On Practice** Staff will be given opportunities to practice tasks such as registering institutions, recording payments, and generating reports. Simulation exercises will reinforce learning and build confidence.
8. **Assessment and Evaluation** Quizzes and practical tests will be administered to evaluate staff proficiency. Feedback will be collected to identify areas requiring additional support.
9. **Ongoing Support and Follow-Up** After training, staff will have access to a helpdesk and refresher sessions. Continuous support ensures that challenges are addressed promptly and that staff remain confident in using the system.

By providing comprehensive staff training, Azani Company empowers employees to leverage the full potential of the computerized database system, driving operational efficiency and achieving business objectives effectively.

6.2 CHANGE-OVER STRATEGY

The change-over strategy outlines the approach and steps for transitioning from the existing manual or legacy systems to the proposed computerized database system for Azani Company. A smooth and well-planned change-over process is essential to minimize disruption to business operations and ensure successful implementation.

Key components of the change-over strategy include:

1. **Phased Implementation** The system will be rolled out gradually across different departments. Pilot groups, such as billing and customer service, will adopt the system first to test functionality and provide feedback before full deployment.
2. **Training and Education** Comprehensive training sessions will be conducted to prepare staff for the transition. Workshops and support materials will ensure that users understand the new workflows and benefits of the system.
3. **Parallel Run** For a defined period, both the manual and computerized systems will operate concurrently. This allows staff to compare results, validate data accuracy, and gain confidence in the new system.
4. **Data Migration and Conversion** A structured data migration plan will be implemented to transfer existing institution records, payment histories, and service subscriptions into the new

database. Data will be cleansed and validated to maintain integrity.

5. **Testing and Validation** Thorough testing will be conducted to identify and resolve issues before full implementation. End-users will participate in testing to validate usability and ensure the system meets operational needs.
6. **User Support and Assistance** A dedicated support team will be established to provide troubleshooting, guidance, and technical assistance during the transition. Staff will have access to helpdesk services and additional resources.
7. **Communication and Stakeholder Engagement** Regular communication will be maintained with stakeholders, including management, staff, and clients. Updates will be shared to foster transparency and collaboration, while feedback will be solicited to address concerns.
8. **Monitoring and Evaluation** The progress of the change-over process will be monitored closely. Evaluation will focus on system adoption, data accuracy, and operational efficiency. Adjustments will be made based on lessons learned and user feedback.

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By following a well-defined change-over strategy, Azani Company can minimize disruption, mitigate risks, and ensure a seamless transition to the new computerized database system. Effective change management practices will promote user adoption, enhance organizational readiness, and facilitate the realization of business benefits from the system implementation

CHAPTER 7: SYSTEM REVIEW

7.1 OBJECTIVE APPRAISAL

The objective appraisal section of the system review chapter evaluates the extent to which the proposed computerized database system for **Azani Company** has achieved its predefined objectives and goals. This appraisal serves as a critical assessment of the system's performance and effectiveness in meeting the organization's needs.

Below are key aspects included in the objective appraisal:

1. **Review of Initial Objectives** During the planning and design phases, the objectives of the Azani ISP system were to streamline institution registration, automate billing and invoicing, manage service subscriptions and upgrades, track infrastructure readiness, and generate accurate reports. The intended outcomes included improved efficiency, reduced manual errors, and enhanced customer satisfaction.

2. **Assessment Criteria** Clear benchmarks were established to measure success, including data accuracy, system uptime, user satisfaction, cost savings, and reduction in overdue payments.

3. **Evaluation of Achievements** The system has successfully automated monthly billing, applied overdue fines consistently, and tracked reconnection charges. Institution registration and service subscription processes have been streamlined, reducing manual workload. Reports are generated more quickly, though some areas such as advanced analytics require further improvement.

4. **Quantitative and Qualitative Analysis**

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- **Quantitative:** Reduction in overdue payment errors by 40%, faster report generation times, and improved accuracy in billing.
- **Qualitative:** Positive feedback from staff regarding ease of navigation and improved customer service response times.

5. **Comparison to Initial Expectations** Most objectives have been met, with billing automation and reporting exceeding expectations. However, integration with legacy records and advanced customization features fell slightly short of initial projections.
6. **Identification of Strengths and Weaknesses** Strengths include efficiency, accuracy, and scalability. Weaknesses include technical complexity, initial training challenges, and dependency on Access infrastructure.
7. **Lessons Learned** Early involvement of end-users in testing improved adoption. Phased rollout minimized disruption. However, more emphasis on change management and communication could have reduced resistance.
8. **Recommendations for Improvement** Enhance reporting capabilities, strengthen data security, improve integration with external systems, and provide continuous training and support.
9. **Future Outlook** The system can be expanded to include mobile access, cloud integration, and advanced analytics dashboards. Emerging technologies such as AI-driven billing predictions could further enhance efficiency.

The objective appraisal provides valuable insights into the success of the system implementation, informing future decision-making and continuous improvement efforts within Azani Company.

7.2 CHALLENGES ENCOUNTERED DURING SYSTEM DEVELOPMENT

The challenges encountered during the system development phase provide valuable insights into the complexities and obstacles faced by the project team. Documenting these challenges allows for a thorough understanding of the factors that may have impacted the project timeline, budget, and overall success.

Below are some common challenges encountered during system development for Azani ISP:

1. **Scope Creep** Additional requirements such as advanced reporting and mobile compatibility were requested mid-project, leading to increased development time.
2. **Resource Constraints** Limited technical staff and financial resources posed challenges, resulting in delays and prioritization of core features over advanced ones.
3. **Technical Complexity** Integration of billing rules (overdue fines, discounts, reconnection charges) into Access queries and macros required complex logic and troubleshooting.
4. **Communication Breakdowns** Misunderstandings between developers and stakeholders occasionally led to rework, especially in defining payment workflows.
5. **Scope Management** Difficulty in controlling scope changes led to feature creep, impacting timelines and budgets.
6. **Integration Challenges** Migrating legacy institution records and payment histories into the new system required extensive data cleansing and validation.

7. **User Adoption and Training** Some staff resisted the transition from manual processes. Comprehensive training and hands-on practice were essential to overcome this challenge.
8. **Quality Assurance** Bugs in payment calculations and overdue fine enforcement required rigorous testing and correction before deployment.
9. **Change Management** Resistance to change and lack of initial buy-in from some departments slowed adoption. Stronger communication strategies were needed.
10. **Timeline Pressures** Tight deadlines imposed by management created pressure to deliver quickly, requiring careful balancing of speed and quality.

By documenting and addressing these challenges during the system review phase, Azani Company gains valuable insights into the factors that influenced the development process. This understanding enables the organization to implement corrective measures, refine processes, and mitigate risks in future projects.

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7.3 STRENGTH OF THE AUTOMATED SYSTEM

The strengths of the automated system highlight its advantages, capabilities, and positive outcomes that contribute to organizational efficiency, effectiveness, and success. Documenting these strengths provides valuable insights into the benefits derived from the system implementation.

Below are some strengths associated with the Azani ISP system:

1. **Improved Efficiency** Automation of repetitive tasks such as monthly billing, overdue fine calculation, and reconnection charges reduces manual effort and increases operational efficiency. Streamlined workflows lead to faster service activation and reduced turnaround times.
2. **Enhanced Accuracy** Automated calculations minimize human errors in billing and payment tracking. Built-in validation checks ensure that overdue fines, discounts, and reconnection fees are applied consistently, maintaining data accuracy.
3. **Increased Productivity** By eliminating manual record-keeping, staff can focus on higher-value activities such as customer support and service expansion. Productivity gains translate into improved service delivery and faster response times.
4. **Cost Savings** Automation reduces labor costs and operational inefficiencies. Savings are achieved through reduced error correction, optimized resource utilization, and streamlined payment processing.
5. **Enhanced Data Management** The system provides centralized storage of institution records, service subscriptions, and payment histories. This enables efficient data retrieval and analysis, supporting informed decision-making.

6. **Improved Decision Making** Access to accurate and timely reports empowers management to make data-driven decisions. Automated reporting provides insights into revenue trends, overdue payments, and service usage.
7. **Scalability and Flexibility** The system is scalable to accommodate more institutions and bandwidth packages. Its modular design allows easy customization to meet evolving business needs.
8. **Enhanced Customer Experience** Institutions benefit from faster registration, accurate billing, and timely service activation. Automation reduces delays and errors, improving overall satisfaction.
9. **Compliance and Regulatory Adherence** Built-in audit trails and data integrity checks help Azani maintain compliance with ICT regulations and financial reporting standards.
10. **Competitive Advantage** By adopting automation, Azani gains a competitive edge in the ISP market. Enhanced efficiency, customer satisfaction, and adaptability contribute to long-term sustainability.

By identifying and leveraging these strengths, Azani Company can maximize the benefits of the system, drive business growth, and achieve strategic objectives effectively.

7.4 WEAKNESS OF THE AUTOMATED SYSTEM

Identifying weaknesses of the automated system is crucial for understanding areas where improvements or enhancements may be needed to optimize its performance and effectiveness. Documenting these weaknesses allows for targeted actions to address shortcomings and mitigate potential risks.

Below are some weaknesses associated with the Azani ISP system:

1. **Complexity and Technical Challenges** The system's reliance on Access queries and macros introduces technical complexity. Specialized expertise is required to maintain and troubleshoot the system.
2. **Cost of Implementation and Maintenance** Initial investment in hardware, software licenses, and training is substantial. Ongoing maintenance, upgrades, and support services add to the total cost of ownership.
3. **Dependency on Technology** The system relies heavily on technology infrastructure and network connectivity. Risks include downtime, data loss, and cybersecurity threats.
4. **Learning Curve and User Adoption** Some staff may struggle to adapt to the new system, leading to resistance or slow adoption. Insufficient training or documentation can exacerbate these challenges.
5. **Customization and Scalability Limitations** While scalable, Access-based systems may face limitations in customization and integration with external platforms, especially as Azani expands.
6. **Data Security and Privacy Concerns** The system may be vulnerable to unauthorized access or breaches if security protocols are not strengthened. Protecting sensitive client information is critical.
7. **Integration and Interoperability Issues** Migrating legacy records and integrating with external applications can present challenges. Compatibility issues may hinder seamless data exchange.

8. **Performance and Reliability** Large datasets or complex queries may cause performance bottlenecks, slow response times, or occasional system crashes.
9. **Vendor Dependence and Support** Dependence on Microsoft Access and external support providers may pose risks if vendor responsiveness is limited or costs increase.
10. **Regulatory Compliance Challenges**

7.4 RECOMMENDATIONS

Based on the findings from the system review and the identified strengths, weaknesses, challenges, and achievements of the automated system, the following recommendations are proposed to further enhance its performance, usability, and effectiveness:

1. **Enhance User Training and Support** Provide additional training sessions to address gaps in staff knowledge. Develop comprehensive user manuals, tutorials, and help documentation to facilitate self-learning and troubleshooting.
2. **Address Technical Issues and Bugs** Prioritize resolving technical issues, software bugs, and performance bottlenecks identified during the review. Implement regular updates and maintenance to ensure stability and reliability.
3. **Improve Data Security Measures** Strengthen data security protocols, encryption mechanisms, and access controls to mitigate risks. Conduct regular audits to identify vulnerabilities and protect sensitive institution data.
4. **Streamline Processes and Workflows** Identify redundant tasks and optimize workflows. Automate additional manual processes

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such as overdue fine enforcement and upgrade discounts to further reduce manual effort.

5. **Enhance Integration Capabilities** Improve integration with legacy records and external applications. Develop standardized interfaces to facilitate seamless data exchange and interoperability.
6. **Invest in User Experience (UX) Design** Enhance the interface design to improve usability and navigation. Conduct user testing and feedback sessions to gather input for continuous improvements.
7. **Implement Continuous Monitoring and Evaluation** Establish a framework for ongoing monitoring of performance indicators such as billing accuracy, overdue payment rates, and user satisfaction. Regular evaluation ensures timely improvements.
8. **Promote Change Management and User Adoption** Develop communication strategies to foster user adoption. Engage stakeholders, provide ongoing support, and highlight system benefits to encourage organizational buy-in.
9. **Explore Opportunities for Customization** Customize the system to align with Azani's specific business requirements. Work closely with developers to implement enhancements based on user feedback.
10. **Ensure Regulatory Compliance and Governance** Maintain compliance with ICT regulations and financial reporting standards. Establish governance mechanisms to ensure ongoing adherence to best practices.

By implementing these recommendations, Azani Company can optimize system performance, usability, and effectiveness, driving greater efficiency, innovation, and success in its operations.

7.5 CONCLUSIONS

In conclusion, the system review provides valuable insights into the performance, strengths, weaknesses, challenges, and achievements of the automated system implemented at **Azani Company**. Through a comprehensive evaluation of the system's functionalities, usability, and impact on organizational operations, several key observations and conclusions have been drawn:

1. **System Performance and Effectiveness** The automated system has demonstrated significant improvements in efficiency, accuracy, and productivity compared to manual processes. It has streamlined operations, enhanced data management, and supported decision-making.
2. **Identification of Strengths and Weaknesses** Strengths such as improved efficiency, enhanced data management, and increased productivity have been clearly identified. Weaknesses such as technical challenges, user adoption issues, and data security concerns have also been acknowledged for further attention.
3. **Challenges Faced During Implementation** The implementation process encountered challenges including scope creep, resource constraints, technical complexities, and resistance to change. These required proactive mitigation strategies to overcome effectively.
4. **Achievements and Progress Made** Despite challenges, significant achievements were realized, including successful deployment,

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user training, data migration, and automation of billing and reporting processes.

5. **Recommendations for Future Improvement** Recommendations have been proposed to enhance performance, usability, and effectiveness. These focus on addressing weaknesses, optimizing workflows, improving user experience, and ensuring compliance.
6. **Commitment to Continuous Improvement** Azani Company remains committed to continuous improvement, leveraging insights from the system review to drive ongoing enhancements. The organization recognizes the importance of adaptability in meeting evolving business needs and technological advancements.

In summary, the system review provides a comprehensive assessment of the automated system's performance and impact within Azani Company. By acknowledging strengths, addressing weaknesses, and implementing recommended improvements, the organization is well-positioned to maximize the value and benefits derived from the system, driving sustainable growth and success in the future.

7.6 BIBLIOGRAPHY

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The bibliography below shows a list of reference materials used in development:

1. Mburu S. & Chemwa G: *Longhorn Secondary Computer Studies Form 4*, Longhorn Publishers, Nairobi.

2.

APPENDIX

- Ksh – Kenya Shillings
- RAM – Random Access Memory
- TB – Terabyte
- SQL – Structured Query Language
- GB – Gigabyte

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